

RASID

Smart Port & Maritime Congestion Predictor

An AIS-Based Predictive Data Engineering and MLOps Pipeline for Egyptian Maritime Gateways

Graduation Project Proposal

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Summary

Rasid (راصد, Arabic for "observer") is a data engineering and MLOps graduation project that forecasts vessel congestion at Egyptian maritime gateways: how many ships will wait at anchorage, and for how long, 6 and 12 hours ahead. The main analysis covers Port Said and the Suez Canal approaches; Alexandria is the live showcase.

This proposal contains the three parts requested by the supervisor:

- **Problem statement** (Section 1): background, the problem, its impact, the research gap, aim and objectives, scope and success criteria.
- **Similar systems** (Section 3): commercial, free, academic and local solutions, a feature comparison, and where Rasid is stronger and weaker.
- **Software requirements** (Section 4): 149 functional, interface, quality, compliance and data requirements, following ISO/IEC/IEEE 29148:2018 [1] and ISO/IEC 25010:2023 [2].

Section 2 explains the proposed solution and its free data sources, and Section 5 gives the team, milestones and risks. The user-interface prototype is being designed in Figma; its screens are listed in Section 4.2.

> **In one sentence**
>
> Rasid is an open, Egypt-focused system that forecasts anchorage queues and waiting times 6 to 12 hours ahead at the Suez Canal approaches and Alexandria, evaluated transparently against baselines and real disruptions.

1. Problem Statement

1.1 Background

The Suez Canal links the Mediterranean Sea at Port Said with the Red Sea at Suez and carries about 12% of global maritime trade, while nearly 90% of Egypt's foreign trade moves through its seaports [3]. After the Red Sea disruption cut canal revenue from $10.2 billion in 2023 to $4 billion in 2024, traffic is now recovering, with transits up 10% in fiscal year 2025/26 [4].

Ships arriving at a port or at the canal often wait at an anchorage, a designated area of sea outside the entrance, until a berth, a pilot or a convoy slot is free. When arrivals exceed capacity, this queue grows: that is congestion. Ships continuously broadcast their position and speed through the Automatic Identification System (AIS), and satellite AIS data now makes these queues measurable hour by hour [5].

1.2 The Problem

> **Problem statement**
>
> Congestion at Egypt's main maritime gateways is detected only after it has formed, because no system turns the available vessel-tracking data into short-term forecasts of anchorage queues and waiting times. Operators cannot plan ahead, and the cost of waiting falls on shipping lines, ports and the wider supply chain.

Rasid addresses this by forecasting, 6 and 12 hours ahead, how many vessels will wait at anchorage and for how long, at Port Said and the Suez Canal approaches (main analysis) and at Alexandria (live showcase). This horizon gives operators time to adjust berth and pilot schedules, advise ships to arrive later, or reschedule onward transport.

1.3 Impact

Unplanned waiting costs every party in the maritime supply chain.

Party	Cost of congestion today	Benefit of a forecast
Port and canal authorities	Pilots, tugs and berths allocated at short notice	Early warning to plan resources
Ship operators and agents	Demurrage (delay penalties), fuel and crew costs while idle	Expected waiting times to plan arrivals
Logistics operators	Missed berth windows; trucks and trains rescheduled	Advance notice to plan onward transport

The 2021 Ever Given grounding shows how fast a queue can build: the canal was blocked from 23 to 29 March [6], and more than 300 ships were soon waiting at anchorage [7].

1.4 Research Gap

- **No forecast exists.** Congestion is managed from declared arrival times and experience, and the team found no public short-term forecast for these gateways. AIS-based congestion prediction has been shown in research [8] but, to the team's knowledge, not applied to Egyptian gateways.
- **The data is fragmented.** Free live AIS feeds do not cover Egyptian waters, satellite AIS arrives about 5 days late [5], and the Port Authority's public ship board is live but partial.
- **Models drift.** Forecasts lose accuracy as traffic patterns shift, as the Red Sea disruption showed [4], unless they are monitored and retrained.

1.5 Aim and Objectives

The aim is to build and evaluate a data platform that forecasts vessel congestion 6 to 12 hours ahead and keeps those forecasts reliable in operation. The objectives are to:

1. Collect and clean free vessel-tracking, port and weather data in a layered data lake.
2. Derive congestion measures (waiting vessels and waiting time) from AIS data, following Peng et al. [8].
3. Train forecasting models for 6-hour and 12-hour horizons that outperform simple baselines.
4. Serve forecasts through an API and a live map dashboard.
5. Keep models reliable through automated monitoring, retraining and rollback.

1.6 Scope

The project must be complete for a defense expected in June 2027, using free public data on a single 16 GB machine.

In scope	Out of scope
Port Said, the Suez Canal approaches and Alexandria	Other ports
Port-level forecasts of waiting vessels and waiting time	Arrival-time prediction for individual ships
Forecasts that inform operators' decisions	Automated scheduling of berths, pilots or convoys
Free public data	Paid data, hardware receivers and satellite radar imagery
Academic, non-commercial use	Commercial use of non-commercially licensed data

1.7 Success Criteria

- **Accuracy:** on unseen time periods, forecast error (MAE and RMSE) at 6 and 12 hours is lower than that of simple baselines, such as assuming the current queue stays the same.
- **Robustness:** forecasts are also evaluated on the 2021 Ever Given disruption, and their error is reported separately from normal periods.

2. Proposed Solution

2.1 Overview

Rasid is one pipeline serving two ports. Vessel positions are clustered (DBSCAN with convex hulls) to find anchorage zones; ships staying in a zone form waiting episodes; hourly counts of waiting ships and their waiting times become the labels [8]. Gradient-boosted models (XGBoost) then forecast the queue and the waiting time for each zone with a range (P10, P50, P90), and every forecast is compared with three baselines and explained with SHAP.

Port	How it is covered
Port Said / Suez Canal approaches (main analysis)	Trained and backtested on satellite AIS history from Global Fishing Watch (hourly, 2017 to about 5 days ago) [5]. The data is not live, so Port Said is demonstrated with a replay simulator, including the 2021 Ever Given disruption.
Alexandria (live showcase)	Live forecasts from the Alexandria Port Authority public ship board [9], polled every 30 minutes, combined with Global Fishing Watch history. Live predictions start after a 4–6 week side-by-side collection of both sources.

2.2 Data Sources

Only free public data is used; paid data and hardware receivers are out of scope.

Source	Port	Role	Notes
Global Fishing Watch AIS vessel presence [5]	Port Said / Suez, Alexandria	Training, labels, backtests	Non-commercial licence (CC BY-NC 4.0); about 5 days late; validated at Port Said (24 of 24 hours covered)
Alexandria Port Authority ship board [9]	Alexandria	Live source and official labels	Official arrival, berthing and departure times; polled politely; permission requested through the supervisor
IMF PortWatch [10]	Suez and ports	Validation	Daily transits and port calls
Suez Canal Authority [11]	Suez	Validation	Official monthly transit totals
Open-Meteo [12]	All	Weather features	Wind, gusts and wave height

Free live AIS networks such as aisstream.io were tested in October 2026 and showed no coverage of Egyptian waters, which is why satellite history and the Port Authority board were chosen.

2.3 Architecture and Technology

Data flows through a layered data lake (Bronze: raw; Silver: cleaned; Gold: model-ready). A streaming path handles the live Alexandria board and a batch path handles history; both use one shared feature library, so training and live forecasts always compute features the same way.

Layer	Technology
Ingestion	Python collectors, one per source → Apache Kafka
Stream processing	Spark Structured Streaming (PySpark)
Storage	MinIO (S3-compatible) for Parquet; PostgreSQL + PostGIS
Orchestration	Apache Airflow
Zones and labels	scikit-learn DBSCAN, shapely, geopandas
Feature store	Feast (offline Parquet, online Redis)
Models	XGBoost quantile models; optional LSTM; SHAP explanations

Layer	Technology
Registry and tracking	MLflow (aliases @champion, @challenger, @previous)
Serving and dashboard	FastAPI; map dashboard; Grafana for operations
Monitoring and quality	Evidently (drift), Great Expectations (data quality)
Infrastructure	Docker Compose with profiles on one 16 GB machine; GitHub Actions CI

3. Similar Systems and Competitor Analysis

3.1 Purpose and Key Findings

This analysis reviews commercial products, free public tools, academic work and local players that address the same problem as Rasid: measuring or forecasting vessel congestion, with a focus on the Suez Canal approaches and Egyptian ports. It answers two questions: where Rasid is stronger than existing solutions, and where they are stronger than Rasid.

- **The idea is not new.** Several commercial platforms already measure and forecast port congestion from AIS data, and Portcast tracks almost the same metrics Rasid plans to forecast [13].
- **The specific combination is not covered.** No reviewed solution forecasts congestion for the Suez Canal approaches or Egyptian ports hours ahead with a published, reproducible evaluation.
- **Suez-focused work exists, but it measures rather than forecasts.** IMF PortWatch publishes daily transit counts [10], Chico et al. measure dwell times during disruptions [14], Lind et al. derive queue positions [15], and Budiman predicts average annual daily traffic [16].
- **Rasid cannot win on data coverage or speed.** Its defensible ground is local specialization, official Alexandria data, transparent evaluation and a complete, deployed pipeline.

3.2 Method

Competitors were found through web searches for port congestion tracking and prediction products, academic sources (arXiv, PLOS ONE, MIT DSpace) and news coverage of the Suez Canal. Each was assessed against the same criteria: coverage, output, forecast horizon, data and latency, Egypt focus, published evaluation, and access (free, paid or research-only).

Limitation: commercial features are taken from public product pages and press releases. The team had no paid access, so vendor claims could not be independently verified.

3.3 Commercial Platforms

Product	What it offers	Relevance to Rasid
Portcast	Port congestion tracker for 600+ ports: average vessels waiting at anchorage per day, median waiting time, a congestion category and a long-tail indicator. The public view is updated weekly; full data and an API are paid [13, 17]	Closest direct competitor: nearly the same metrics
Kpler (including MarineTraffic)	Container and terminal congestion intelligence. Holds patent EP3879468 (originally filed by Jakota Cruise Systems), which forecasts terminal congestion up to 14 days ahead from AIS and itinerary data [18, 19]	Strongest data network; terminal-level, days-ahead forecasts
Weathernews — Berth Waiting Forecast	Launched in February 2024. Combines AIS from about 70,000 ships with weather data to show real-time delay status per port and congestion forecasts up to one week ahead, in ships per day [20]	Closest method (AIS, weather and machine learning), but global and berth-oriented

Product	What it offers	Relevance to Rasid
Lloyd's List Intelligence	Predictive analytics for arrival, berthing and departure times and congestion levels; documents AIS data-quality problems such as missing ETAs [21]	Vessel-centric; confirms a data risk Rasid also faces
Signal Ocean	Port congestion dashboard and API for dry bulk and tanker vessels, with ten years of history [22]	Different vessel segments; much deeper history
Windward	Maritime AI platform with predicted ETAs and congestion levels at container ports, built on more than 12 years of proprietary data [23]	Container- and ETA-focused
Visibility platforms (project44, FourKites, SeaVantage)	Congestion is one feature in broad supply-chain visibility suites; SeaVantage markets terminal forecasts 48 to 72 hours ahead (vendor claim) [24]	Congestion is a side feature, not the product

3.4 Free Public Tools

Tool	What it offers	Relevance to Rasid
IMF PortWatch	Free platform from the IMF and the University of Oxford. Publishes daily AIS-derived port calls and chokepoint transits, including the Suez Canal, with downloads and an API [10]	Free and covers Suez, but counts transits after the fact; it does not forecast queues or waiting times. Rasid uses it for validation

3.5 Academic and Research Work

Work	What it does	Relevance to Rasid
Peng et al. (2023)	Deep learning approach to estimate and predict port congestion from AIS data [8]	The method Rasid's congestion labels follow; not Suez-specific
Chico et al. (2025), PLOS ONE	Open-source AIS framework measuring vessel counts, dwell and idle times during waterway disruptions; applied to the 2021 Suez blockage using the North Anchorage near Port Said, the Great Bitter Lake and the South Anchorage near Suez [14]	Closest academic work on the Suez anchorages; measures, does not forecast
Lind et al. (2021)	Showed that canal passage after the Ever Given reopening was mostly first-come, first-served, with deviations caused by convoys; proposed AIS-based queue numbers to predict waiting [15]	Supports convoy-aware modelling; a proposal, not a system
Budiman (2024), MIT thesis	Predicts average annual daily ship traffic in the Suez Canal: satellite imagery plus AIS with an LSTM (13.2% MAPE) and a discrete-event simulation with an LSTM (3.8% MAPE) [16]	Same region, different target: yearly average traffic, not short-term waiting
Xue et al. (2026), AIS-TGNN	Predicts congestion escalation with a temporal graph attention network and generates explanations with a language model; six months of AIS data at Los Angeles and Long Beach (test AUC 0.761) [25]	Stronger explainability; different port; research prototype

3.6 Local Landscape (Egypt)

- **Suez Canal Authority:** publishes official navigation statistics [11] and controls convoy scheduling, so it holds convoy data Rasid does not have. It is both a reference point and a potential validation partner.
- **Alexandria Port Authority:** publishes a public ship board with arrival, berthing and departure times, but no forecasts [9]. It is Rasid's live data source.

- **Shipping agents:** local agencies such as Leth Agencies have published manual updates on vessels waiting at Port Said, the Great Bitter Lake and Suez [26]. They are potential users who currently track queues by hand.
- **Egyptian startups:** the search found no provider of AIS-based congestion forecasting for Port Said, Suez or Alexandria. The search covered public web sources only.

3.7 Feature Comparison

The matrix compares Rasid with the closest competitor in each group. “Not public” means the information is not available from public sources.

Aspect	Portcast	Kpler	Weathernews	IMF PortWatch	Chico et al.	Rasid
Coverage	600+ ports	Global terminals	Global ports	Global ports and chokepoints	Suez, Bosporus, Bab el-Mandeb	**Port Said and Suez Canal approaches; Alexandria**
Main output	Vessels waiting, median waiting time, category	Terminal congestion	Ships waiting per day	Daily transits and port calls	Vessel counts, dwell and idle times	**Vessels waiting and waiting time**
Forecasts ahead	Yes (paid)	Up to 14 days	Up to 1 week	No	No	**6 and 12 hours**
Data freshness	Public view weekly	Own AIS network	Real time	Daily, several days' lag	Historical study	**Alexandria every 30 min; Port Said by historical replay (satellite data about 5 days late)**
Egypt-specific	No	No	No	Partly (Suez transits)	Yes (Suez case study)	**Yes**
Official local data	No	No	No	No	Validated with SCA statistics	**Yes (Alexandria Port Authority board)**
Weather inputs	Not public	Not public	Yes	No	No	**Yes**
Published accuracy	No	No	No	Method published	Yes	**Yes (baselines and backtests)**
Explainability	Not public	Not public	Not public	Not applicable	Not applicable	**SHAP feature attributions**
Access	Paid (free weekly snapshot)	Paid	Paid	Free	Free (open code)	**Academic, non-commercial**

3.8 Where Rasid Is Stronger

Rasid's advantages come from depth on one region and from transparency, not from data scale.

#	Advantage	Stronger than	Why it holds
1	Focus on Egyptian gateways	All global platforms	Rasid models the actual Suez anchorage zones (such as the North Anchorage near Port Said) derived from AIS, rather than treating Port Said as one port among hundreds
2	Official local data	All reviewed competitors	Alexandria forecasts use the Port Authority's own ship board, a source none of the reviewed solutions uses
3	Short-term forecasts of waiting	IMF PortWatch and all academic work	Rasid predicts waiting vessels and waiting time 6 and 12 hours ahead; the free and academic tools only measure what already happened
4	Published, reproducible evaluation	All commercial platforms	Results are reported against three baselines on time-based backtests, including the 2021 Ever Given disruption; no commercial platform reviewed publishes verifiable accuracy
5	Complete, deployed system	All academic work	Rasid delivers a running service: prediction API, map dashboard, data-quality checks, drift monitoring, retraining and rollback. The academic works are studies or prototypes

#	Advantage	Stronger than	Why it holds
6	Open method on free data	All commercial platforms	Anyone can reproduce the results from free sources; commercial methods are proprietary

A conditional advantage: convoy-aware modelling. Waiting at the canal is shaped by convoy scheduling [15], and no reviewed competitor advertises convoy-aware forecasting. Rasid can claim this only once it includes convoy-related features inferred from AIS data and shows that they improve accuracy.

3.9 Where Competitors Are Stronger

The established players lead on data, scale and maturity; Rasid addresses each gap by design rather than by competing head-on.

Area	Competitor advantage	How Rasid responds
Data freshness	Kpler and Windward run their own AIS networks with continuous updates [18, 23]	Satellite AIS for Port Said arrives about 5 days late, so Port Said forecasts are demonstrated by historical replay; live forecasting runs at Alexandria on the official board. This limit is stated openly
Coverage	Hundreds of ports worldwide [13, 18]	Deliberate choice of depth over breadth: two Egyptian gateways
History	Signal Ocean has ten years of history [22]; Windward more than 12 years [23]	Satellite AIS from 2017 onward gives enough history to train and backtest
Forecast horizon	Kpler forecasts up to 14 days [19]; Weathernews up to one week [20]	Rasid targets the operational window of hours; longer horizons are out of scope
Convoy data	The Suez Canal Authority holds actual convoy schedules	Infer convoy patterns from AIS; seek the Authority as a validation partner
Explainability	AIS-TGNN generates natural-language explanations [25]	SHAP feature attributions per forecast
Maturity and customers	Established products with paying customers	Rasid is an academic prototype; its value is evaluation quality, not market share
Commercial use	Competitors sell their data and forecasts	Satellite AIS from Global Fishing Watch is licensed for non-commercial use only, so Rasid stays academic; any commercial version would need licensed data

3.10 Positioning and Implications

> **Positioning**
>
> Rasid is an open, Egypt-focused system that forecasts anchorage queues and waiting times 6 to 12 hours ahead at the Suez Canal approaches and Alexandria, evaluated transparently against baselines and real disruptions. It does not compete with global platforms on coverage; it competes on local depth and verifiable accuracy.

For this positioning to hold at the defense, the project must deliver the following:

1. Prove the forecast beats the baselines, and publish the results, including the Ever Given backtest.
2. Build and test convoy-related features inferred from AIS data; claim convoy-aware modelling only if they improve accuracy.
3. Complete the Alexandria overlap study comparing the Port Authority board with satellite AIS, so the official-data advantage is backed by evidence.
4. Approach the Suez Canal Authority for validation data on convoys and transits.

5. Keep commercial claims (pricing, customers) out of academic deliverables while the data licence is non-commercial.

4. Software Requirements Specification

The requirements follow the structure of ISO/IEC/IEEE 29148:2018 [1]; quality requirements are grouped by the nine characteristics of ISO/IEC 25010:2023 [2]. Each requirement has a unique ID, a priority and a verification method.

Legend	Meaning
Priority (MoSCoW)	M = Must (needed for the defense), S = Should, C = Could
Verify	T = Test, D = Demonstration, I = Inspection, A = Analysis
Trace	S1–S9 = planned screens (Section 4.2); D-/P-/O- = project decisions; C- = constraints; G1/M1–M5 = milestones

4.1 Users and Roles

User class	Role	Needs	Characteristics
Shipping agent / freight forwarder	Viewer or Analyst	Know whether a ship will wait and for how long; avoid demurrage; warn clients early.	Business user, frequent use, limited data skills, Arabic or English.
Terminal / port operator	Analyst	Plan berths, cranes and labour before the queue forms; set alerts.	Operational user, daily use, works under time pressure.
Carrier operations	Viewer	Adjust ship speed or call order to avoid idle waiting.	Uses the API or dashboard occasionally.
Port-authority analyst	Analyst	Review history, replay events, compare with official figures.	Analytical, infrequent but deep use.
ML / data engineer (team)	Administrator	Monitor data sources and models, approve zones, promote or roll back models.	Technical expert, daily use during operation.
Supervisor / examiner	Viewer	Evaluate results, backtests and the MLOps loop.	Academic reviewer, occasional use.

4.2 Planned Screens (UI Prototype)

The prototype is being designed in Figma. These screen IDs are used for traceability and will be replaced by the final frame names.

Screen	Content
S1 Sign-in & port selector	E-mail/password sign-in, port choice, language switch.
S2 Live map	Anchorage zones, waiting ships coloured by waiting time, congestion level, data time.
S3 Forecast	Actual queue and 6 h / 12 h forecast with P10–P90 band; expected waiting time; table view.
S4 Forecast explanation	Top factors (SHAP) written in plain language.
S5 Vessel queue	Table of waiting ships with filters and CSV export.
S6 Alerts	Alert rules, channels, history of delivered alerts.
S7 History & replay	Period picker, replay controls, forecast-versus-actual comparison, backtest metrics.
S8 Model health (admin)	Active model, accuracy versus baselines, drift, promote/rollback with reason, event history, zone approval.
S9 Data sources (admin)	Each source's last update, expected delay, health, data-quality results.

4.3 Functional Requirements

M1 — Data Ingestion

ID	Requirement	Priority	Trace	Verify
FR-ING-01	The system shall poll the Alexandria Port Authority ship board every 30 minutes and retrieve the expected, in-port and departed ship lists.	M	D-008, S9	T
FR-ING-02	The poller shall wait at least 2 seconds between consecutive requests and identify the project in its User-Agent header.	M	C-03	I
FR-ING-03	The system shall store every board snapshot unchanged in Bronze, partitioned by date and hour, with the retrieval time in UTC.	M	C-06	T
FR-ING-04	The system shall compare consecutive snapshots and publish change events (newly expected, arrived, berthed, berth shift, departed) to Kafka.	M	Arch. §2	T
FR-ING-05	The system shall convert all Port Authority times from Cairo local time (daylight-saving aware) to UTC on ingest.	M	DR-01	T
FR-ING-06	The system shall download Global Fishing Watch hourly vessel presence daily for each configured area (Port Said anchorage, Suez Bay, Alexandria) and store it in Bronze.	M	D-008	T
FR-ING-07	The system shall backfill Port Said history from 2017 to the latest available day; an interrupted backfill shall resume where it stopped.	M	G1	D
FR-ING-08	The system shall download IMF PortWatch daily series once a day for validation.	S	D-008	T
FR-ING-09	The system shall import Suez Canal Authority monthly transit totals (manual CSV import is acceptable).	C	D-008	I
FR-ING-10	The system shall retrieve hourly wind speed, gusts and wave height from Open-Meteo for each zone centroid.	S	Features	T
FR-ING-11	On a failed request, a collector shall retry with exponential backoff, log the failure and continue at the next schedule without losing stored data.	M	NFR-RE-03	T
FR-ING-12	Re-running ingestion for a time window already collected shall not create duplicate records.	M	—	T
FR-ING-13	The replay simulator shall re-stream any stored Bronze time window into Kafka in original order at a configurable speed from 1× to 60×.	M	Demo, S7	D
FR-ING-14	The replay simulator shall optionally inject faults (wrong speed units, missing hours) for drift and resilience demonstrations.	S	M3	D

M2 — Data Processing & Quality

ID	Requirement	Priority	Trace	Verify
FR-DQ-01	The system shall clean AIS records: drop impossible positions (latitude 91, longitude 181, out of range), set sentinel values to null (speed 102.3, heading 511, course 360) and remove duplicates on vessel, time and position.	M	Data notes §2	T
FR-DQ-02	The system shall drop a position whose implied speed from the previous position exceeds 50 knots.	S	Data notes §2	T
FR-DQ-03	Records failing validation shall be moved to a quarantine table with the reason, never silently deleted.	M	—	T

ID	Requirement	Priority	Trace	Verify
FR-DQ-04	The system shall maintain a vessel registry (IMO, MMSI, name, type, size) and enrich coarse vessel types from the registry and from the Port Authority board where IMO numbers match.	S	GFW report	T
FR-DQ-05	The system shall run data-quality suites on every Silver and Gold write; a failing suite shall block the write and raise an alert.	M	S9	T
FR-DQ-06	The system shall store the result of every data-quality run for later review.	S	S9	I
FR-DQ-07	Port Authority records open for more than 60 days shall be flagged as suspected legacy records and excluded from live features.	S	DQ report	T

M3 — Zones & Labels

ID	Requirement	Priority	Trace	Verify
FR-LBL-01	The system shall derive anchorage and berth zones from stationary vessel positions using DBSCAN clustering and convex or concave hulls.	M	Peng 2023	T
FR-LBL-02	Zones shall be stored as versioned GeoJSON and become active only after an administrator approves them on the map.	M	S8	D
FR-LBL-03	The system shall build anchorage episodes per vessel: start = inside an anchorage zone below 1 knot for at least 30 minutes; end = leaves the zone above 2 knots or crosses a gate line; waiting time = end − start.	M	Data notes §3	T
FR-LBL-04	Episodes shorter than 1 hour, longer than 14 days, or containing data gaps over 6 hours shall be flagged, not deleted.	M	Data notes §3	T
FR-LBL-05	The system shall compute hourly zone tables: occupancy, arrivals, departures, median completed wait and median wait by arrival hour.	M	S2, S3	T
FR-LBL-06	The system shall assign each zone-hour a congestion level (LOW / NORMAL / HIGH / SEVERE) from the occupancy percentile over the trailing 90 days.	M	S2, S3	T
FR-LBL-07	Only matured labels (default maturity lag 7 days, configurable) shall be used for training and evaluation.	M	Data notes §3	T
FR-LBL-08	For Alexandria, labels shall come from official Port Authority arrival, berthing and departure times reconstructed from snapshot changes.	S	D-008	T
FR-LBL-09	Activating a new zone version shall trigger a rebuild of all dependent labels and record the new label version.	S	—	T

M4 — Feature Management

ID	Requirement	Priority	Trace	Verify
FR-FEA-01	Every feature transformation shall be defined once in the shared feature library and imported by both the batch and the streaming pipelines.	M	P-001, C-05	I
FR-FEA-02	The system shall compute the v1 feature catalog: queue state, arrivals and departures over 1/3/6/24 h, approach pressure, calendar (including Egyptian holidays and Ramadan), weather and target lags.	M	Data notes §4	T
FR-FEA-03	Features shall be served through a feature store with point-in-time correct retrieval for training and an online store for live inference.	M	Arch. §2	T
FR-FEA-04	Each online feature shall have a time-to-live; a feature older than its TTL shall be treated as stale.	M	FR-API-03	T
FR-FEA-05	An automated parity test shall replay a fixed 24-hour fixture through both pipelines and fail the build if the features differ beyond tolerance.	M	M2	T

M5 — Forecasting & Model Training

ID	Requirement	Priority	Trace	Verify
FR-ML-01	The system shall compute and report three baselines for every target and horizon: persistence, seasonal-naive (same hour last week) and the convoy-schedule heuristic.	M	P-005, C-07	T
FR-ML-02	The system shall train one XGBoost model per horizon producing P10, P50 and P90 forecasts of anchorage occupancy and expected waiting time.	M	M1	T
FR-ML-03	Forecast horizons shall be 6 h and 12 h, with 24 h as an option, configurable without code changes.	M	O-002	T
FR-ML-04	Models shall be evaluated only with rolling-origin, time-based backtests, reporting per horizon and per traffic regime: MAE, RMSE, pinball loss, skill versus persistence, P10–P90 coverage and lead time for HIGH/SEVERE onsets.	M	C-07	A
FR-ML-05	Every trained model shall be registered with its metrics, training-data window, zone version, label version and code version.	M	MA-04	T
FR-ML-06	The system shall compute SHAP values giving the top factors behind each forecast.	S	S4	T
FR-ML-07	An LSTM model may be trained and shall be kept only if it beats XGBoost on the same backtest folds.	C	—	A
FR-ML-08	Training and evaluation datasets shall contain only real observed data, never synthetic data.	M	C-06	I

M6 — Forecast Serving (API)

ID	Requirement	Priority	Trace	Verify
FR-API-01	The system shall expose a versioned REST API that lists zones and returns, per zone and horizon, P10/P50/P90 occupancy, expected wait and congestion level.	M	§3.1.4	T
FR-API-02	Every forecast response shall include the issue time, the model version, the data freshness in seconds and a stale flag.	M	IF-UI-03	T
FR-API-03	When input features are stale, the API shall return stale = true and shall not present the forecast as current.	M	NFR-SA-02	T
FR-API-04	The API shall return the top contributing factors for each forecast.	S	S4	T
FR-API-05	Every prediction shall be logged with zone, horizon, quantiles, model version and issue time, for later evaluation and audit.	M	DR-04	T
FR-API-06	The serving layer shall load the @champion model and switch to a new champion without downtime when the alias changes.	M	P-008	T
FR-API-07	The API shall expose health, readiness and metrics endpoints.	M	§3.1.4	T
FR-API-08	The API shall publish an OpenAPI description.	S	NFR-CO-02	I
FR-API-09	Forecast endpoints shall require a valid user session or API key.	S	NFR-SE-04	T

M7 — Dashboard

ID	Requirement	Priority	Trace	Verify
FR-UI-01	The user shall be able to select a port (Port Said / Suez or Alexandria) and a zone.	M	S1	D

ID	Requirement	Priority	Trace	Verify
FR-UI-02	The dashboard shall show a map of zones with the vessels currently waiting, coloured and labelled by congestion level, and the time of the latest data.	M	S2	D
FR-UI-03	For a selected zone, the dashboard shall show the recent actual queue and the 6 h / 12 h forecast with its P10–P90 range.	M	S3	D
FR-UI-04	The dashboard shall explain a selected forecast in plain language using its top factors.	S	S4	D
FR-UI-05	The dashboard shall list waiting vessels with name, type, zone, arrival time and elapsed wait, with sorting and filtering.	S	S5	D
FR-UI-06	Analysts shall be able to export the vessel queue and forecast tables to CSV.	C	S5	T
FR-UI-07	The dashboard shall replay a past period (e.g., the 2021 Ever Given blockage) and compare forecasts with what actually happened.	S	S7	D
FR-UI-08	The dashboard shall be available in Arabic and English, with a right-to-left layout in Arabic.	S	S1–S9	D
FR-UI-09	The Global Fishing Watch attribution shall be shown wherever its data is displayed.	M	C-02	I

M8 — Alerts

ID	Requirement	Priority	Trace	Verify
FR-ALR-01	Analysts shall be able to create alert rules per zone on forecast thresholds (e.g., forecast queue above N ships within 12 h, or level HIGH or worse).	S	S6	T
FR-ALR-02	Alerts shall be delivered in the dashboard and by e-mail within 5 minutes of the forecast that triggers them.	S	S6	T
FR-ALR-03	The same alert shall not be repeated within a configurable cool-down period (default 3 hours).	S	S6	T
FR-ALR-04	Administrators shall be alerted when a data source delivers nothing within its expected interval or when a data-quality suite fails.	M	S9	T

M9 — MLOps Monitoring & Model Lifecycle

ID	Requirement	Priority	Trace	Verify
FR-OPS-01	The system shall check input drift daily on key features and trigger retraining when drift exceeds the configured threshold.	M	P-002	T
FR-OPS-02	The system shall compare the champion's error on matured labels with @previous and with persistence over a rolling 7-day window.	M	P-002	T
FR-OPS-03	When the champion performs worse than @previous or persistence over that window, the system shall roll back automatically by moving @champion to @previous and alert the administrators.	M	M3	D
FR-OPS-04	The system shall retrain weekly and register the result as @challenger.	M	P-002	T
FR-OPS-05	A challenger shall run in shadow mode (predictions logged, not shown) and be promoted only if it beats the champion on matured labels.	S	P-002	T
FR-OPS-06	Administrators shall be able to promote or roll back a model manually; a reason is mandatory and recorded in the audit log.	M	S8	D

ID	Requirement	Priority	Trace	Verify
FR-OPS-07	The model-health screen shall show the active model version, accuracy versus baselines, drift status and the history of retrain, promote and rollback events.	M	S8	D
FR-OPS-08	The data-sources screen shall show each source's last successful update, its expected delay (e.g., Global Fishing Watch ≈ 5 days) and its health status.	M	S9	D
FR-OPS-09	The system shall generate a model card for each champion (training window, metrics, known limitations).	C	S8	I

M10 — Users & Access

ID	Requirement	Priority	Trace	Verify
FR-USR-01	The system shall support three roles: Viewer (dashboards), Analyst (Viewer + alerts, exports, replay) and Administrator (Analyst + zones, models, data sources, users).	M	S1	T
FR-USR-02	Users shall sign in with e-mail and password; administrators shall be able to create, disable and assign roles to accounts.	M	S1	T
FR-USR-03	Users shall be able to reset a forgotten password by e-mail.	C	S1	T
FR-USR-04	Sessions shall expire after a configurable period of inactivity (default 8 hours).	S	NFR-SE	T

4.4 External Interface Requirements

User Interfaces

ID	Requirement	Priority	Trace	Verify
IF-UI-01	The dashboard shall be a responsive web application usable from 1024 px width, with read-only views usable at 768 px (tablet).	S	S1–S9	T
IF-UI-02	Every screen shall show a header with the Rasid logo, the selected port, the user's role and a language switch (Arabic / English).	S	S1–S9	I
IF-UI-03	Every data panel shall show the time of its latest data and a visible "stale" marker when the data is older than its expected update interval.	M	S2–S9	D
IF-UI-04	Forecast charts shall draw actual values as a solid line, forecasts as a dashed line and the P10–P90 range as a shaded band, with a table view of the same values.	M	S3	I
IF-UI-05	Colour shall never be the only way information is shown; congestion levels also use text labels.	S	S2, S3	I

Software Interfaces

ID	Requirement	Priority	Trace	Verify
IF-SW-01	Port Authority board: HTTPS web extraction of the expected, in-port and departed lists with full TLS certificate verification.	M	D-008	T
IF-SW-02	Global Fishing Watch 4Wings API: hourly presence per vessel at high spatial resolution for named areas, authenticated with a token read from the environment.	M	D-008	T
IF-SW-03	IMF PortWatch API: daily port-call and chokepoint-transit series.	S	D-008	T

ID	Requirement	Priority	Trace	Verify
IF-SW-04	Open-Meteo API: hourly forecast and marine variables for zone centroids.	S	D-008	T
IF-SW-05	Apache Kafka topics for raw events, cleaned events and predictions; every message carries a source field and an event time.	M	Arch. §2	T
IF-SW-06	Object storage through the S3 API (MinIO) for Bronze, Silver and Gold Parquet files.	M	P-004	I
IF-SW-07	PostgreSQL + PostGIS for zones, vessel registry, episodes, users, alerts and the prediction log.	M	P-004	I
IF-SW-08	MLflow model registry using aliases @champion, @challenger and @previous.	M	P-008	T
IF-SW-09	Feast feature store with an offline store (Parquet) and an online store (Redis).	M	Arch. §2	T

Communication Interfaces

ID	Requirement	Priority	Trace	Verify
IF-CM-01	All external and user-facing traffic shall use HTTPS (TLS 1.2 or later).	M	—	T
IF-CM-02	The forecast API shall exchange JSON over REST and be versioned under /v1.	M	Arch. §5	T
IF-CM-03	Alert e-mails shall be sent through an SMTP service configured by the administrator.	S	S6	T

Forecast API endpoints

Endpoint	Returns
GET /v1/zones	List of zones: id, name, type (anchorage / berth), port, geometry reference.
GET /v1/zones/{zone_id}/forecast?horizons=6,12	Issue time, model version, data freshness (s), stale flag, and per horizon: occupancy P10/P50/P90, expected wait P50/P90, congestion level, top drivers.
GET /v1/zones/{zone_id}/history?from=&to=	Hourly actual occupancy and waits with the forecasts issued for the same hours.
GET /v1/vessels/{mmsi}/status	Current episode and elapsed wait for one vessel (later phase).
GET /healthz · /readyz · /metrics	Liveness, readiness and Prometheus metrics.

4.5 Non-Functional (Quality) Requirements

Numeric targets are agreed project targets and are checked with the verification method given.

Functional suitability

ID	Requirement (measurable target)	Priority	Verification
NFR-FS-01	At the 6-hour horizon the champion model shall beat persistence on MAE (skill > 0) in rolling-origin backtests for Port Said. If it does not by gate G2 (end of Nov 2026), the target or horizon is revised.	M	A: backtest report
NFR-FS-02	The P10–P90 range shall contain the actual value in 70–90% of matured cases (target 80%).	S	A: coverage per horizon
NFR-FS-03	For HIGH/SEVERE congestion onsets the system shall report precision, recall and median warning time; target median warning time ≥ 6 hours.	S	A: event backtest

ID	Requirement (measurable target)	Priority	Verification
NFR-FS-04	Daily vessel counts derived by Rasid shall be compared weekly with IMF PortWatch and SCA figures, and the deviation reported.	S	A: validation report

Performance efficiency

ID	Requirement (measurable target)	Priority	Verification
NFR-PE-01	Forecast API: 95% of requests answered within 500 ms with 20 concurrent users on the reference machine.	S	T: load test
NFR-PE-02	Dashboard: first load within 3 s on a 10 Mbit/s connection; map and chart interactions respond within 1 s.	S	T: browser timing
NFR-PE-03	Freshness: within 5 minutes of each Port Authority poll, Alexandria features and forecasts shall be updated.	M	T: timestamps
NFR-PE-04	The daily batch run (download, Silver/Gold update, features) shall finish within 2 hours.	S	T: Airflow logs
NFR-PE-05	The complete stack shall run within 16 GB RAM using Compose profiles; the demo set (core, stream, ml, ui) shall use at most 12 GB.	M	T: docker stats
NFR-PE-06	The streaming path shall sustain 10× the normal event rate, using the replay simulator on real data.	S	T: load test

Compatibility

ID	Requirement (measurable target)	Priority	Verification
NFR-CO-01	The dashboard shall work on the two latest versions of Chrome, Edge and Firefox.	M	T: browser matrix
NFR-CO-02	The API shall use JSON over HTTPS and publish an OpenAPI 3 description.	S	I
NFR-CO-03	Storage shall use the S3 API so that moving to a cloud object store is a configuration change.	S	I
NFR-CO-04	Geospatial data shall use WGS84 (EPSG:4326) and GeoJSON.	M	I

Interaction capability (usability)

ID	Requirement (measurable target)	Priority	Verification
NFR-IC-01	A first-time user shall find the 12-hour forecast for a chosen port within 2 minutes without training (at least 4 of 5 test users).	S	T: usability test
NFR-IC-02	Forecasts and levels shall be expressed in plain language with units (ships, hours); maritime terms shall have a tooltip explanation.	S	I
NFR-IC-03	Accessibility: WCAG 2.2 level AA — text contrast at least 4.5:1, full keyboard operation, visible focus, charts readable without colour.	S	I + T: audit tool
NFR-IC-04	The interface shall respect the operating system’s reduced-motion setting.	C	I
NFR-IC-05	Error messages shall state what happened and what the user can do next.	C	I

Reliability

ID	Requirement (measurable target)	Priority	Verification
NFR-RE-01	The board poller shall complete at least 99% of its scheduled polls per calendar month, excluding outages of the source itself.	M	A: poll log
NFR-RE-02	Every gap in stored data longer than 1 hour shall be detected and recorded with its cause.	M	T
NFR-RE-03	After a crash or restart of any service the system shall resume without data loss or manual repair (Bronze is immutable; Kafka keeps at least 7 days).	M	T: kill-and-restart test
NFR-RE-04	When a source fails, the dashboard shall keep working and show the last good data marked as stale.	M	T: fault injection
NFR-RE-05	The forecast API shall be available at least 99% of the time during pilot and demo periods.	S	A: uptime monitor
NFR-RE-06	The relational database and model registry shall be backed up daily, and a restore shall be tested before the defense.	S	T: restore drill

Security

ID	Requirement (measurable target)	Priority	Verification
NFR-SE-01	Secrets (API tokens, passwords) shall be kept only in environment files excluded from version control, and never written to code or logs.	M	I: repo scan
NFR-SE-02	All external connections shall use TLS 1.2 or later with certificate verification.	M	T
NFR-SE-03	Passwords shall be stored with a slow salted hash (bcrypt or Argon2).	M	I
NFR-SE-04	Role-based access control shall be enforced on the server for every endpoint and administrative action.	M	T: role tests
NFR-SE-05	Administrative actions (model promote/rollback, zone approval, user changes) shall be written to an append-only audit log with user, time and reason.	M	T
NFR-SE-06	Dependencies shall be scanned for known vulnerabilities in CI; critical findings block a release.	S	T: CI
NFR-SE-07	After 10 failed sign-in attempts an account shall be locked for 15 minutes.	S	T

Maintainability

ID	Requirement (measurable target)	Priority	Verification
NFR-MA-01	Each data source shall be an independent collector adapter; adding a source shall need no change downstream.	M	I
NFR-MA-02	Automated tests shall cover at least 70% of the lines in the feature library and the labelling modules.	S	T: coverage report
NFR-MA-03	Every change shall pass CI (lint, unit tests, parity test) before it is merged.	M	I: CI history
NFR-MA-04	Zones, labels, features, models and data windows shall be versioned so that any past forecast can be reproduced.	M	D: reproduce a forecast
NFR-MA-05	Code shall follow the project conventions (Python 3.11, ruff, type hints on public functions).	S	I

Flexibility (portability and scalability)

ID	Requirement (measurable target)	Priority	Verification
NFR-FL-01	Adding a new port shall require configuration only (areas, zones, source settings), not code changes.	S	D
NFR-FL-02	The system shall install on a clean machine with one documented command in under 30 minutes, excluding downloads.	S	T
NFR-FL-03	A live AIS feed shall be addable later as a new adapter without changes to features, models or the API.	S	I
NFR-FL-04	Moving to a cloud deployment shall require only configuration changes (storage endpoint, credentials).	C	A

Safety

ID	Requirement (measurable target)	Priority	Verification
NFR-SA-01	Rasid is decision support only: every forecast screen shall state that forecasts are estimates and must not be used for navigation.	M	I
NFR-SA-02	A stale or failed forecast shall never be shown as current (see FR-API-03).	M	T
NFR-SA-03	No zone version shall become active without human approval, because zones affect all labels (see FR-LBL-02).	M	D

4.6 Compliance and Data Requirements

Compliance

ID	Requirement (measurable target)	Priority	Verification
LC-01	Global Fishing Watch data shall be used for non-commercial research only, with the attribution “Powered by Global Fishing Watch” (CC BY-NC 4.0).	M	I
LC-02	The Port Authority board shall be polled politely (C-03); raw board data shall never be published; written permission shall be requested through the supervisor before any public demonstration.	M	I
LC-03	Open-Meteo data shall be used under its non-commercial terms.	M	I
LC-04	User personal data (name, e-mail) shall be collected only for accounts, not shared, and deleted on request, in line with Egypt’s Personal Data Protection Law (Law No. 151 of 2020).	S	I
LC-05	The public repository shall contain documents only until the supervisor approves (D-009).	M	I

Data

ID	Requirement (measurable target)	Priority	Verification
DR-01	Times shall be stored in UTC (ISO-8601), coordinates in WGS84, speed in knots and distance in nautical miles.	M	I
DR-02	Bronze data shall be immutable and kept indefinitely; Silver and Gold shall be kept and versioned.	M	I
DR-03	Data shall be partitioned by date=YYYY-MM-DD/hour=HH and stored as Parquet with zstd compression.	S	I

ID	Requirement (measurable target)	Priority	Verification
DR-04	Every forecast shall be stored with zone, horizon, quantiles, model version and issue time; actual values are joined when they mature.	M	T
DR-05	Synthetic data, if used for load tests, shall be stored separately and labelled as synthetic.	M	I

4.7 Design Constraints

ID	Constraint
C-01	Free data only: no paid data and no hardware (team decision, 6 Oct 2026; D-008).
C-02	Global Fishing Watch data is non-commercial (CC BY-NC 4.0) and must carry the attribution “Powered by Global Fishing Watch”.
C-03	Port Authority board: at least 2 seconds between requests, at most one poll every 30 minutes, project named in the User-Agent, raw data never published.
C-04	The whole stack must run under Docker Compose on one 16 GB machine using profiles.
C-05	Feature logic lives only in the shared feature library and is imported by both batch and streaming jobs (P-001).
C-06	No synthetic data in training or evaluation; synthetic data only for load tests.
C-07	Time-based splits only; every model is reported against the baselines.
C-08	The public GitHub repository holds documents only until the supervisor approves (D-009).
C-09	Satellite imagery (SAR) is out of scope.
C-10	Python 3.11, ruff, pytest, type hints on public functions; Conventional Commits.

4.8 Verification Scenarios

These scenarios are run before the defense and recorded:

Scenario	What it proves
V-01 Ever Given replay	Replaying 9 Mar – 12 Apr 2021 at 60× speed, the forecast rises before the queue does; backtest metrics against all baselines (FR-ML-01/04, FR-UI-07, NFR-FS-01/03).
V-02 Live Alexandria day	Polls every 30 minutes, events, features and forecasts update within 5 minutes; freshness shown on every panel (FR-ING-01..05, NFR-PE-03, IF-UI-03).
V-03 Drift injection	Replaying with corrupted speed units raises a drift alert and triggers retraining (FR-ING-14, FR-OPS-01).
V-04 Bad model rollback	Deploying a deliberately worse model is detected on matured labels and rolled back automatically (FR-OPS-02/03).
V-05 Source outage	Stopping a collector raises an admin alert; the dashboard shows stale data instead of failing (FR-ALR-04, NFR-RE-04, NFR-SA-02).
V-06 Parity test	Batch and streaming features are equal on the 24-hour fixture in CI (FR-FEA-05).
V-07 Usability test	Five users complete the core tasks; times and errors recorded (NFR-IC-01).
V-08 Load and restart	10× replay load, API latency, kill-and-restart without data loss (NFR-PE-01/06, NFR-RE-03).

5. Team and Plan

5.1 Team

Member	Role	Responsibility
Mohamed Mahmoud	Team Lead — MLOps Engineer & Data Engineer	MLOps and platform; integration; supervisor liaison
Reem Adel	Data Engineer	Data pipeline workstream (to be assigned)
Alhassan Mohamed	Data Engineer	Data pipeline workstream (to be assigned)
Youssef Nasser	Data Engineer	Data pipeline workstream (to be assigned)
Youssef Elkomy	Data Engineer	Data pipeline workstream (to be assigned)

5.2 Milestones

The team builds a thin, working end-to-end slice first, then adds the streaming path, the MLOps loop and the final evaluation.

ID	Target date	Milestone
G1	19 Oct 2026	Board poller and daily satellite-AIS download running 24/7; Port Said history backfilled; prediction target fixed
M1	29 Nov 2026	Thin slice on real data (Port Said): zones → labels → features → model beats baselines → API → map
M2	24 Jan 2027	Streaming features (Alexandria live); batch/stream feature parity test passing in CI; overlap study done
M3	21 Mar 2027	Closed MLOps loop demonstrated (drift → retrain, regression → rollback) and SHAP explanations
M4	9 May 2027	Feature freeze; backtests including Ever Given; usability test
M5	June 2027	Thesis and defense (date to be confirmed)

5.3 Key Risks

Risk	Mitigation
Live data depends on one public web page (the Port Authority board)	Polite polling, every snapshot stored immediately, alerts on layout changes, written permission sought through the supervisor; Port Said analysis does not depend on the board
Satellite AIS arrives about 5 days late and is non-commercial only	Port Said is analysed and backtested rather than predicted live; the live demo runs at Alexandria
The board and AIS describe Alexandria differently	4–6 weeks of side-by-side collection before live predictions; features both sources can produce
The model may not beat the baselines	Gate at M1: if it fails at 6 hours, revise the target or horizon before building more infrastructure
Traffic patterns keep shifting (Red Sea recovery)	Drift monitoring, weekly retraining and metrics reported per traffic regime

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Appendix A — Glossary

Term	Meaning (plain language)
AIS	Automatic Identification System: radio messages every large ship broadcasts with its identity, position, speed and heading.
Anchorage	A sea area outside a port or canal where ships wait at anchor until a berth or a canal convoy is available.

Term	Meaning (plain language)
Berth	A place at a quay where a ship is tied up to load or unload cargo.
Convoy	A group of ships the Suez Canal Authority schedules to transit the canal together.
Demurrage	A daily penalty paid to the ship owner when a ship is held longer than the time agreed in its contract.
ETA	Estimated time of arrival.
Episode	One continuous stay of one ship in an anchorage, from arrival to departure; its length is the waiting time.
Occupancy / queue	The number of ships waiting in an anchorage zone at a given hour.
Congestion level	LOW / NORMAL / HIGH / SEVERE, from the occupancy compared with the previous 90 days.
Horizon	How far ahead a forecast looks (6 h, 12 h, optionally 24 h).
P10 / P50 / P90	Forecast range: a 10% chance the value is below P10, 50% below P50 (the median), 90% below P90.
Baseline	A simple reference forecast (e.g., “the queue stays as it is now”) that the model must beat.
SHAP	A method that shows which inputs pushed a forecast up or down.
Drift	A change in the incoming data compared with the data the model was trained on.
Champion / challenger	The model currently serving forecasts / a newly trained candidate tested in the background.
Rollback	Switching back to the previous model when the current one performs worse.
Bronze / Silver / Gold	Data layers: raw as received / cleaned / aggregated and ready for models.
Feature / feature store	A model input (e.g., arrivals in the last 6 h) / the system that stores features consistently for training and live use.
TTL	Time-to-live: how long a stored value is considered fresh.
GFW	Global Fishing Watch: free satellite-AIS vessel-presence data for non-commercial research (CC BY-NC 4.0).
APA board	Alexandria Port Authority public ship board: official expected, in-port and departed ships.
SCA	Suez Canal Authority.
MMSI / IMO	Ship radio identifier (9 digits) / permanent hull number (7 digits).
MoSCoW	Priority scale: Must, Should, Could, Won’t (this release).